Introduction to Translational Lung Research – Fundamentals of Lung Biology and Physiology

Monday, October 19, 2026

Location: Seminar room 002, Helmholtz Pioneer Campus (HPC, Building 3630), Helmholtz Munich

12:00 – 12:30	Arrival and Registration
12:30 – 13:00	Welcome and Introduction Dr. Doreen Franke, Dr. Melanie Penning Prof. Dr. A. Önder Yildirim Dr. Malte Lücken & DZL DataLung Steering Board Prof. Dr. Martijn Nawijn & RESPIRE-EXCEL Board
13:00 – 13:45	Lung Structure Prof. Dr. Markus Diefenbacher (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
13:45 – 14:15	Lung Structural Cells I: Alveolar and Bronchial Epithelial Cells, Stem Cells Dr. Roxana Wasnick (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
14:15 – 14:45	Lung Structural Cells II: Endothelial Cells and Fibroblasts Dr. Gerald Burgstaller (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
14:45 – 15:15	Break
15:15 – 15:45	Introduction to Immunology Dr. Aicha Jeridi (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
15:45 – 16:15	Lung Immune Cells I: Monocytes, Macrophages, DCs, Neutrophils, Innate Lymphoid Cells, NK Cells Dr. Chiara Goracci (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
16:15 – 16:45	Lung Immune Cells II: T and B Lymphocytes Dr. Tom Conlon (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
16:45 – 17:15	Inter-Organ Communication Dr. Roxana Wasnick (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)

Tuesday, October 20, 2026

Location: Seminar room 002, Helmholtz Pioneer Campus (HPC, Building 3630), Helmholtz Munich

08:55 – 09:00	Welcome to the Day
09:00 – 09:30	Molecular Pathways involved in Human Trained Immunity Prof. Dr. Christian Lindermayr (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
09:30 – 10:00	Aging Dr. Maria Camila Melo Narvaez (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)

10:00 – 10:30	Lung Organotypic Models (PCLS, Organoids, ALI) Dr. Maria Camila Melo Narvaez (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
10:30 – 11:00	iPSCs as a Model of Lung Disease Camille Barro (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
11:00 – 11:30	Break
11:30 – 12:30	Lung Function + Lung Function Tests Dr. Jannik Ruwisch (Department of Respiratory Medicine and Infectious Diseases, Hannover Medical School, BREATH)
12:30 – 16:00	Interactive Lecture: PROM (Patient Reported Outcome Measures) in Lung Health (**Please bring your laptops!**), *incl. group work & lunch break* PD Dr. Jan Fuge (Data Management, Hannover Medical School, BREATH)
16:00 – 16:45	Introduction to Epidemiology - Focus on Air Pollution and Lung Disease PD Dr. Stefan Karrasch (LMU Klinikum, CPC-M)
16:45 – 17:15	Environmental Challenges for the Lung Dr. Tobias Stöger (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
17:15 – 18:00	Keynote Lecture: *State or Fate: Genes, Environment and Programming in Early Origins of Disease* Prof. Dr. Markus Weckmann (University Hospital Schleswig-Holstein, ARCN) Chairs: Annalea Brüggemann, Carina Pawlow (BREATH)

Location: “Walk of Creativity”, Helmholtz Pioneer Campus (HPC, Building 3630, 2nd Floor)

18:00 – 21:00	Joint DataLung-RESPIRE-EXCEL-EHLRS **Pizza-Get-together**

Wednesday, October 21, 2026

Location: Seminar room 002, Helmholtz Pioneer Campus (HPC, Building 3630), Helmholtz Munich

08:55 – 09:00	Welcome to the Day
09:00 – 09:30	Lung Development + BPD PD Dr. Kai Förster (LMU Klinikum, Center for Comprehensive Developmental Care, CPC-M)
09:30 – 10:15	Asthma and Allergies Dr. Nicole Maison (Dr. von Haunersches Kinderspital, LMU Klinikum, CPC-M)
10:15 – 10:45	Chronic Obstructive Pulmonary Disease (COPD) Dr. Aicha Jeridi (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
10:45 – 11:15	Break
11:15 – 11:45	Idiopathic Pulmonary Fibrosis (IPF) Dr. Diana Porras-Gonzalez (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)

11:45 – 12:15	Cystic Fibrosis and Bronchiectasis Dr. Jeremias Götschke (LMU Klinikum, CPC-M)
12:15 – 13:00	Lung Cancer Prof. Dr. Markus Diefenbacher (Institute of Lung Health and Immunity, Helmholtz Munich, LMU, CPC-M)
13:00 – 14:00	Lunch Break
14:00 – 14:30	Pulmonary Hypertension Dr. Stefan Hadzic (Excellence Cluster Cardio-Pulmonary Institute, UGMLC)
14:30 – 15:00	Pneumonia, Acute Respiratory Distress Syndrome, Acute Lung Injury Dr. Christina Malainou (Department of Medicine V Internal Medicine, Infectious Diseases and Infection Control, University Hospital Giessen and Marburg, Excellence Cluster Cardio-Pulmonary Institute, UGMLC)
15:00 – 15:30	Lung Viral Infection Prof. Dr. Heiko Adler (Walther-Straub-Institute of Pharmacology and Toxicology/ LMU, CPC-M)
15:30 – 16:00	Break
16:00 – 17:00	Clinical Diagnosis of Lung Diseases Lung Transplantation Dr. Carlo Mümmler (Institute of Lung Health and Immunity, Helmholtz Munich, LMU Klinikum, CPC-M)
18:00	Optional: **Dinner @ Andechser Am Dom** (Frauenplatz 7, 80333 München)
20:00	Optional: Social Evening (When registering, please select your preferred option)

Thursday, October 22, 2026

Location: Room 003, Building 31, Helmholtz Munich

09:10 – 09:15	Welcome to the Day
09:15 – 10:15	Single Cell Transcriptomics Dr. Malte Lücken (Institutes of Lung Health and Immunity, and Computational Biology, Helmholtz Munich, CPC-M), Dr. Svenja Gaedcke (Bioinformatics Platform, BREATH)
10:15 – 11:15	Omics in Lung Research – Data Analysis and Integration Dr. Malte Lücken (Institutes of Lung Health and Immunity, and Computational Biology, Helmholtz Munich, CPC-M), Dr. Svenja Gaedcke (Bioinformatics Platform, BREATH)
11:15 – 11:45	Break
11:45 – 12:15	Imaging in Lung Research: Confocal, Light Sheet; 3D Reconstructions, Data Analysis Dr. Gerald Burgstaller (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
12:15 – 12:45	*In Vivo* Imaging Prof. Dr. Markus Rehberg (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
12:45 – 13:30	Digital Image Processing Dr. Urs Eisenmann (Institute of Medical Informatics, TLRC)

13:30 – 19:00	Munich Sightseeing (Guided “Altstadt Tour” in English, kindly sponsored by the DZL Central Office; **meeting point @ Marienplatz - Mariensäule, 03:50 pm**)
19:00	Joint DataLung-RESPIRE-EXCEL-EHLRS **Social and Dinner @ Hacker-Pschorr Donisl** (Marienplatz: Weinstraße 1, 80333 München) Due to the number of guests, and to help speed up the ordering process at the restaurant, we kindly ask that you select your menu in advance (ideally by October 20): Donisl Menu 22.10.2026 https://www.donisl.com/fileadmin/user_upload/unsere-kueche/Speisekarte/September_Karte_-2.pdf, English: https://www.donisl.com/fileadmin/user_upload/unsere-kueche/Speisekarte/Sommerkarte__Englisch_.pdf

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Friday, October 23, 2026

Location: Seminar room 002, Helmholtz Pioneer Campus (HPC, Building 3630), Helmholtz Munich

08:55 – 09:00	Welcome to the Day
09:00 – 09:30	Metagenomics Dr. Ilona Rosenboom (Research Core Unit Genomics, BREATH)
09:30 – 10:15	What you need to know about Genome-Wide Association Studies (GWAS) and Polygenic Scores (PGS) Prof. Dr. Inke König (Institute of Medical Biometry and Statistics/ University of Lübeck, University Medical Center Schleswig-Holstein, ARCN)
10:15 – 10:45	Break
10:45 – 11:30	Aerosol Deposition in the Lung and (Pre)clinical Inhalation Techniques Dr. Otmar Schmid (Institute of Lung Health and Immunity, Helmholtz Munich, CPC-M)
11:30 – 12:15	Introduction to Drug Development and Delivery Dr. Benjamin Winkeljann (Department of Pharmacy/ LMU, CPC-M)
12:15 – 13:00	Feedback and Farewell Dr. Malte Lücken, Dr. Doreen Franke, Dr. Melanie Penning

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This *Lung Science Training* is part of the curriculum for all doctoral researchers (class of 2026) enrolled in the Environmental Health and Lung Research School (EHLRS), the Core Area "Lung Health and Immunity" within the Munich Medical Research School (MMRS) at the Faculty of Medicine, Ludwig Maximilian University of Munich (LMU), and for all DZL DataLung School members (German Center for Lung Research, DZL). Please attend this course in person.

This training is jointly organized by the following entities:

Environmental Health and Lung Research School (EHLRS), Helmholtz Munich

Program Directors:

Dr. Melanie Penning
Dr. Doreen Franke

Scientific Board:

Prof. Dr. Markus Diefenbacher
Dr. Aicha Jeridi
Dr. Theodoros Kapellos
Dr. Malte Lücken
Dr. Roxana Wasnick

Director, Institute of Lung Health and Immunity (Helmholtz Munich), and CPC-M:

Prof. Dr. A. Önder Yildirim

DZL DataLung Steering Board, German Center for Lung Research (DZL)

Chair:

Dr. Malte Lücken (CPC-M - Comprehensive Pneumology Center Munich)

Co-Chair:

Dr. Svenja Gaedcke (BREATH - Biomedical Research in Endstage And ObsTructive Lung Disease Hannover)

Program Manager:

Dr. Doreen Franke (CPC-M, DZL Academy)

Steering Board Members:

Dr. Urs Eisenmann (TLRC - Translational Lung Research Center Heidelberg)
PD Dr. Jan Fuge (BREATH)
Prof. Dr. Inke König (ARCN - Airway Research Center North)
Dr. Raphael Majeed (UGMLC - Universities of Giessen and Marburg Lung Center)
Prof. Dr. Rory Morty (DZL Academy, TLRC)

Please do not hesitate to contact us if you have any questions: datalungschool@dzl.de

RESPIRE-EXCEL (Respiratory Precision Medicine PhD Training Network – Excellence through Systems Biology, Spatial and Single-cell Transcriptomics; Horizon Europe Marie Skłodowska-Curie Actions Doctoral Network)

Prof. Dr. Markus Weckmann (University Hospital Schleswig-Holstein, DE)
Prof. Dr. Martijn Nawijn (UMC Groningen, NL)
Dr. Malte Lücken (Helmholtz Munich, DE)

Registration

Please kindly register for the event here: https://webext.mh-hannover.de/soscisurvey/datalungkickoff/

Venue

Helmholtz Munich
Ingolstädter Landstr. 1
85764 Neuherberg

Travel and Accommodation

Arrangements for travel and accommodation need to be made individually, and expenses are to be covered by the sites/supervisors.

When looking for accommodation, consider the area of Garching-Hochbrück — it offers direct bus access to the Helmholtz campus and U-Bahn connections to the city center.

How to get to the Helmholtz Munich campus:

From Hauptbahnhof: Take the **U-Bahn U2** (direction *Feldmoching*), exit at *Am Hart*, then take the **bus 294** to *Garching Hochbrück* or the **bus 295** to *Oberschleißheim*, exit at *Helmholtz Zentrum*

From Garching-Hochbrück: Take the **bus 294** to *Am Hart*, exit at *Helmholtz Zentrum*

From Autobahn A99: Take the exit *München-Neuherberg* from the Autobahn and follow the B13 in the direction of *München / M.-Neuherberg / Arena / Helmholtz Zentrum*. Continue on Ingolstädter Landstraße (B13). Then, turn left onto the campus entrance of Helmholtz Munich. Please use the Visitor Car Park at the entrance.

Please be aware that you will need to register at the entrance of Helmholtz Munich. You should plan at least 5-10 minutes for the walk from the entrance to the seminar rooms.

Meals

Unfortunately, meals will not be provided during the event. However, there are both a small and a larger cafeteria on campus where you can find coffee, snacks, and lunch options.

We're looking forward to spending some informal time together as well! On Tuesday evening, we'll have a casual get-together with pizza and drinks, and on Wednesday and Thursday evenings, we'll enjoy dinner together at a local restaurant - please note that these meals will be self-paid.

Pictures and Videos

Photos and videos will be taken during the event. By attending this event, you consent to being photographed and/or recorded, and grant the DZL the right to use these images and videos for promotional purposes, including but not limited to social media, websites, newsletters, and brochures. If you have any questions or concerns, please feel free to contact us.